Exercise 3 : SQL Case Statements.

Table . products.

product_id	product_name	price.
1	Laptop	1200.00
2	Phone	800
3	Keyboard	45
4	Monitor	300.00
5	Mouse	25

Classify each product by price.

Expensive if price > 1000
Mid-rang if price between 100 and 1000
Budget if price < 100

```
SELECT product_name, price,
    CASE WHEN price < 100 THEN 'budget'
         WHEN price BETWEEN 100 AND 1000 THEN 'mid-rang
         WHEN price > 1000 THEN 'expensive'
    END AS price_category
FROM products;
```

product_name	price	price_category
Laptop	1200.00	expensive
Phone	800.00	mid-range
Keyboard	45.00	budget
Monitor	300.00	midrange
Mouse	25.00	budget

Table: Orders

order_id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

Label each order:

'High value' for orders ≥ 1000

'Medium value' for 500 - 999.99

'Low value' for orders < 500

Expected outcome:

customer_name

amount

order_value_category

```
SELECT customer_name,
       amount,
  CASE WHEN amount ≥ 1000 THEN 'High value'
       WHEN amount BETWEEN 500 AND 99999 THEN
       'Medium value'
       WHEN amount < 500 THEN 'Low value'
  END AS order_value_category
FROM orders;
```

customer name	amount	order_value_category
Alice	150	low value
Bob	560	medium value
Charlie	99999	medium value
Diana	45.50	low value
Ethan	1200	high value

Table : employees.

emp_id	emp_name	department	salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

Categorize employee position.
If in 'IT' and salary > 80000 → 'Senior IT'
If in 'HR' and salary > 55000 → 'Experienced HR'
Otherwise → 'staff'

Expected outcome column
emp_name
department
salary
position_level

```
SELECT  emp_name,
        department,
        salary,
  CASE WHEN  department = 'IT' AND salary > 80000
      THEN  (Senior IT)
      WHEN  department = 'HR' AND salary > 55000
        THEN (experienced HR)
  ELSE  'staff'
END AS 'position_level';
```

emp_name	department	salary	position_level
John	IT	85000	Senior IT
~~Sara~~	HR	60000	Experienced HR
Mark	IT	75000	staff
~~Lucy~~	~~Finance~~	~~95000~~	staff
Tom	HR	55000	staff

Table: students

student_id	student-name	score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

Assign a letter grade

≥ 90 'A'
80 - 89 'B'
70 - 79 'C'
60 - 69 'D'
< 60 'F'

Expected outcome columns:

student_name
score
grade

```
SELECT student_name,
       score,
   CASE WHEN score ≥ 90 THEN 'A'
        WHEN score BETWEEN 80 AND 89 THEN 'B'
        WHEN score BETWEEN 70 AND 79 THEN 'C'
        WHEN score BETWEEN 60 AND 69 THEN 'D'
   ELSE 'F'
   END AS grade;
```

~~Student_name~~ ~~score~~

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

Table deliveries

delivery-id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

Label delivery performance

≤ 30mins 'fast'
31 - 60 mins 'Ontime'
60mins 'Late'

Expected outcome

delivery-id
delivery-time-minutes
performance

```
SELECT delivery-id,
       delivery-time-minutes,
  CASE WHEN delivery-time-minutes ≤ 30 THEN 'Fast'
       WHEN delivery-time-minutes BETWEEN 31
       AND 60 THEN 'On time'
  ELSE 'Late'
  END AS PERFORMANCE
  FROM deliveries;
```

delivery-id	delivery_time-minutes	performance
1	45	ONTime
2	80	Late
3	30	Fast
4	65	Late
5	100	Late.

Table: tickets

tickets_id	issue_type	priority
1	login issue	1
2	server down	3
3	slow system	2
4	email error	2
5	password reset	1

Convert priority to lables:

3 → 'High'
2 → 'Medium'
1 → 'Low'

Expected output columns
issue_type
priority
priority-label

```
SELECT issue-type,
       priority,
  CASE WHEN priority = 3 THEN 'high'
       WHEN priority = 2 THEN 'Medium'
       WHEN priority = 1 THEN 'Low'
END AS priority-label
FROM tickets;
```

issue-type	priority	priority-label
login issue	1	low
server down	3	high
slow system	2	medium
email error	2	medium
password reset	1	low

Table: attendance

Student-id	days-present	total-days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

Calculate attendance % and classify:

≥ 90% → 'Excellent'

75-89% → 'Good'

< 75% → 'needs improvement'

Expected output columns:

student_id

attendance-percentage

attendance-status

```
SELECT    student_id,
          days-present / totaldays * 100 AS attendance status
CASE WHEN (days_present/totaldays*100) ≥75 THEN 'needs
          improvement'
     WHEN (days-present/totaldays*100) BETWEEN 75
          AND 89  THEN 'Good'
     ELSE 'Excellent'
  END AS attendance-status
FROM attendance;
```

student_id	attendence_percentage	attendance status
1	90	excellent
2	60	need improvement
3	96	excellent
4	50	needs improvement
5	100	excellent

Table: products_inventory

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

Label stock status:
0 → 'Out of stock'
1-5 → 'Low stock'
5 → 'In stock'

Expected outcome columns:
product_id
stock_qty
stock_status:

```
SELECT product_id,
       stock_qty,
   CASE WHEN stock = 0 THEN 'out of stock'
        WHEN stock BETWEEN 1 AND 5 THEN 'low stock'
        ELSE 'in stock'
END AS stock_status
FROM products_inventory
```

product_id	stock_qty	stock_status:
1	5	in stock
2	0	out of stock
3	25	in stock
4	10	in stock
5	3	low stock

Table: classes.

class-id	subject	enrolled-students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

Classify by size

≥ 25 → 'Large'
10-24 → 'Medium'
<10 → 'Small'

Expected outcome output:
subject
enrolled_students.
class_size_category

```
SELECT      subject,
            enrolled_students
    CASE WHEN   enrolled students ≥ 25 THEN 'large'
         WHEN   enrolled-students BETWEEN 10 AND
                24   THEN 'medium'
         ELSE   'small'
    END AS  'class-size-category'
FROM classes;
```

subject	enrolled_students	class-size-category
Math	30	Large
English	25	Large
Science	15	medium
Art	5	small
History	20	medium

Payments Table

payments

payment_id	amount	payment_method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	PayPal
5	300.00	Cash.

Apply discount flag:

If payment_method = 'Cash' and amount ≥ 200 → 'Eligible for discount'.
Otherwise → 'Not eligible'

Expected output columns
payment_id
payment_method
amount
discount eligibility

```
SELECT payment_id,
       payment_method,
       amount,
  CASE WHEN payment_method = 'Cash' AND amount
       ≥ 200 THEN 'eligible for discount'
  ~~ELWENE~~
  ELSE 'Not eligible
END AS discount_eligibility
FROM payment;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50	Not eligible
2	Cash	200	eligible for discoum
3	Card	150	not eligible
4	PayPal	75	not eligible
5	Cash	300	eligible